

JEE MAIN REVISION MASTER: ELECTROSTATICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 270 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Electrostatics - Core Application	270 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Electrostatics - Core Application (Very High Yield — 270 PYQs)

Core Principle: Key problem pattern in Electrostatics.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two point charges $q_1 = 3\mu\text{C}$ and $q_2 = -4\mu\text{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ res..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In electrostatics, a co..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular a..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Two point charges $q_1 = 3\mu\text{C}$ and $q_2 = -4\mu\text{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ respectively. Force on charge q_2 is ____ N. (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI Units)

- (A) $(12\hat{i} + 24\hat{j} + 24\hat{k}) \times 10^{-3}$
- (B) $(4\hat{i} + 8\hat{j} + 8\hat{k}) \times 10^{-3}$ **[Correct]**
- (C) $(3\hat{i} + 6\hat{j} + 6\hat{k}) \times 10^{-3}$
- (D) $(-4\hat{i} - 8\hat{j} - 8\hat{k}) \times 10^{-3}$

Correct Answer: B

Solution Analysis: Coulomb's Law states that the force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them. In vector form, the force $\vec{F}_{21}$ exerted on charge q_2 by charge q_1 is: $\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}|^2} \vec{r}$ Where, $\vec{r} = \vec{r}_2 - \vec{r}_1$ is the position vector from q_1 to q_2 , $\hat{r} = \frac{\vec{r}}{|\vec{r}|}$ is the unit vector and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ Substituting the unit vector, we get: $\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}|^2} \vec{r}$